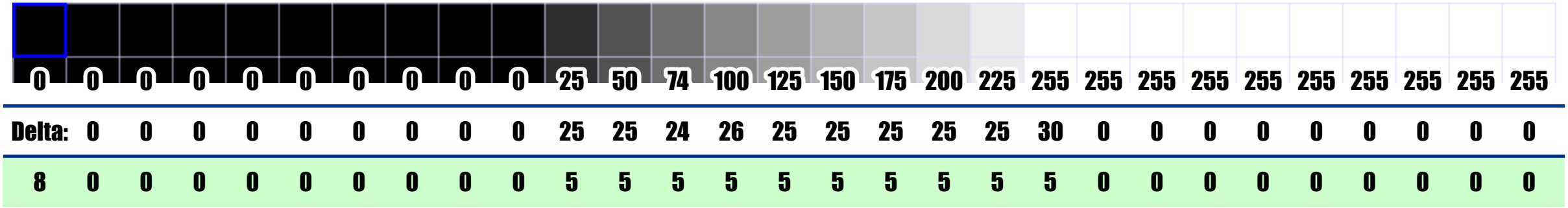


Dmitry Muravyev

Delta encoding

Sprint 1: Compression methods

Delta encoding



	0	0	0	0	0	0	0	0	0	0	25	50	74	100	125	150	175	200	225	255	255	255	255	255	255	255	255	255
Delta:	0	0	0	0	0	0	0	0	0	0	25	25	24	26	25	25	25	25	25	30	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	5	5	5	5	5	5	5	5	5	5	0	0	0	0	0	0	0	0

$8 + 0 \times 9 + 5 \times 10 + 0 \times 9 = 58$ bits

58 vs 232 bits

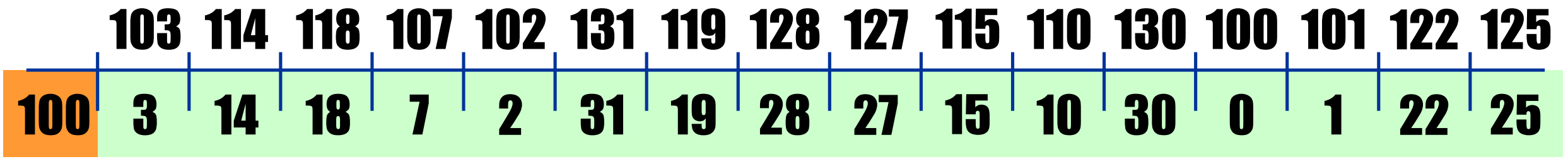
PROFIT: **174** bits (**21.75** bytes)

(theoretical)

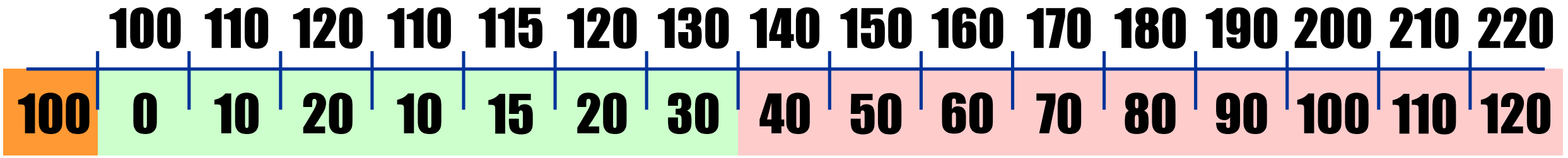
Window bits	Max eff
7	12.5%
6	25.0%
5	37.5%
4	50.0%
3	62.5%
2	75.0%
1	87.5%
0	100%

Fixed window

Window width: 5 bits (0~31)
Window start: X_{min}



Block length: 16



Block length: 7

Floating window

Window width: 5 bits (-16 ~ +15)
Window center: X_{n-1}

103	114	118	107	102	131	119	128	127	115	110	130	100	101	122	125
103	+11	+4	-11	-5	+29	-12	+9	-1	-12	-5	+20	-30	+1	+21	+3

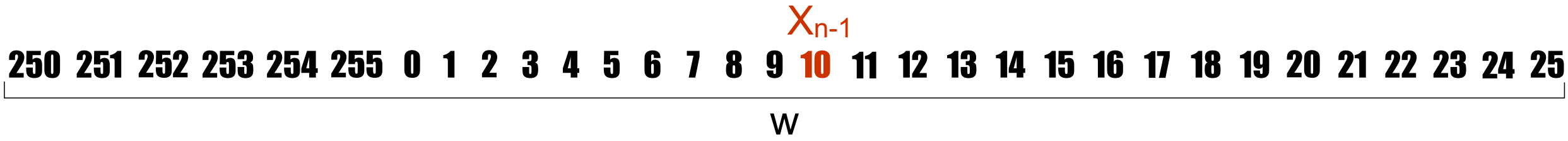
Block length: 5

100	110	120	110	115	120	130	140	150	160	170	180	190	200	210	220
100	+10	+10	-10	+5	+5	+10	+10	+10	+10	+10	+10	+10	+10	+10	+10

Block length: 16

0-Crossing Floating window

Window width: 5 bits (-16 ~ +15)
Window center: X_{n-1}

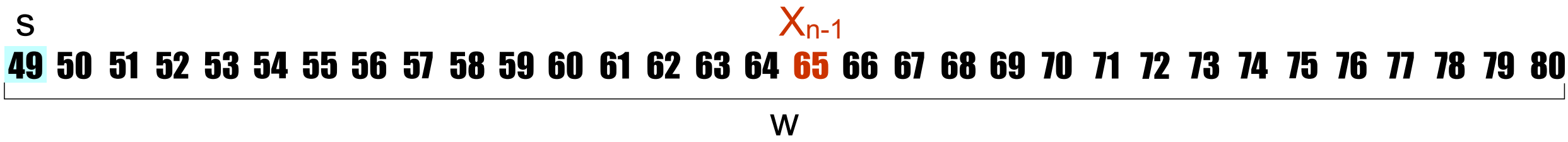


0	2	5	8	10	15	16	0	255	252	250	5	20	30	40	50
0	+2	+3	+3	+2	+5	+1	-16	-1	-3	-2	+11	+15	+10	+10	+10

Block length: 16

Adaptive floating window

Window width (w): 5 bits (0~31)
Window start (s): $X_{n-1} - 2^{w-1}$



100	110	120	110	115	120	130	140	150	160	170	180	190	200	210	220
100	26	26	6	21	21	26	26	26	26	26	26	26	26	26	26
s:	84	94	104	94	99	104	114	124	134	144	154	164	174	184	194

Block length: 16

Adaptive floating window

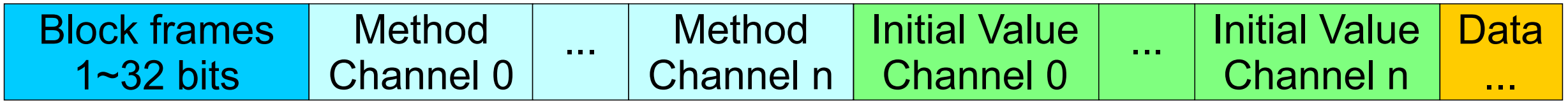
Window width (w): 5 bits (0~31)
Window start (s): $X_{n-1} - 2^{w-1}$

$$\begin{cases} s < 0 \rightarrow s = 0, \\ s + 2^w > N_p \rightarrow s = N_p - 2^w \\ (\text{byte: } N_p = 2^8 = 256) \end{cases}$$

3	14	18	7	2	31	19	28	27	15	10	30	0	1	22	25
3	14	18	5	2	31	4	25	15	3	10	30	-14	1	22	18
s:	0	0	2	0	0	15	3	12	11	0	0	14	0	0	7

Block length: 12

Header



Methods (m):

0 All frames are the same, almost 100% compression (no compressed data). Initial value = first frame.

1~k Fixed Window. Width (w) = m. **k = bits per channel**. Compression per frame: k - w bits. Initial value = minimum.

k No compression, **no Initial value**.

m > k Adaptive Floating Window. Width (w) = m - k. Compression per frame: k - w bits. Initial value = first frame.

m = 0 ~ 2k - 1
Method declaration (bits) = Number_of_the_most_significant_bit(2k-1) + 1

Efficiency

$E_{block} = (1 - B / R_{block}) \times 100\%,$
$E_{overall} = (1 - C / R) \times 100\%,$

- B — Block size, B = Header size + Method (m) × Block frames,
- R_{block} — Raw (uncompressed) data size per block,
- R — Overall raw data size,
- C — Compressed data size (sum of all block sizes).

* All sizes are in bits.

Results

Logo 240x240 Grayscale
Raw bytes: 57600
Max Block Size: 64 bytes
Header bits: 10



Fixed Window:
Method: 0, Compressed bytes: 14283.5, Efficiency: 75.20%
Method: 1, Compressed bytes: 19647.25, Efficiency: 65.89%
Method: 2, Compressed bytes: 24496.75, Efficiency: 57.47%
Method: 3, Compressed bytes: 30206.5, Efficiency: 47.56%
Method: 4, Compressed bytes: 36300, Efficiency: 36.98%
Method: 5, Compressed bytes: 42590.125, Efficiency: 26.06%
Method: 6, Compressed bytes: 48365.25, Efficiency: 16.03%
Method: 7, Compressed bytes: 54274, Efficiency: 5.77%
Method: 8, Compressed bytes: 58725, Efficiency: -1.95%

Floating Window:
Method: 1, Compressed bytes: 19845.625, Efficiency: 65.55%
Method: 2, Compressed bytes: 23664, Efficiency: 58.92%
Method: 3, Compressed bytes: 29105.625, Efficiency: 49.47%
Method: 4, Compressed bytes: 35384, Efficiency: 38.57%
Method: 5, Compressed bytes: 41743.75, Efficiency: 27.53%
Method: 6, Compressed bytes: 47900.25, Efficiency: 16.84%
Method: 7, Compressed bytes: 53614, Efficiency: 6.92%

0-Crossing Floating Window:
Method: 1, Compressed bytes: 19845.625, Efficiency: 65.55%
Method: 2, Compressed bytes: 23664, Efficiency: 58.92%
Method: 3, Compressed bytes: 29105.625, Efficiency: 49.47%
Method: 4, Compressed bytes: 35384, Efficiency: 38.57%
Method: 5, Compressed bytes: 41743.75, Efficiency: 27.53%
Method: 6, Compressed bytes: 47900.25, Efficiency: 16.84%
Method: 7, Compressed bytes: 53102.125, Efficiency: 7.81%

Adaptive Floating Window:
Method: 1, Compressed bytes: 19845.625, Efficiency: 65.55%
Method: 2, Compressed bytes: 23632.25, Efficiency: 58.97%
Method: 3, Compressed bytes: 29031.875, Efficiency: 49.60%
Method: 4, Compressed bytes: 35326.25, Efficiency: 38.67%
Method: 5, Compressed bytes: 41655.25, Efficiency: 27.68%
Method: 6, Compressed bytes: 47457.25, Efficiency: 17.61%
Method: 7, Compressed bytes: 53243.25, Efficiency: 7.56%

Results

Mattress 240x240 Grayscale
Raw bytes: 57600
Max Block Size: 32 bytes
Header bits: 9



Fixed Window:
Method: 0, Compressed bytes: 37989.375, Efficiency: 34.05%
Method: 1, Compressed bytes: 41978, Efficiency: 27.12%
Method: 2, Compressed bytes: 44280.125, Efficiency: 23.12%
Method: 3, Compressed bytes: 43105, Efficiency: 25.16%
Method: 4, Compressed bytes: 44647.5, Efficiency: 22.49%
Method: 5, Compressed bytes: 47102.75, Efficiency: 18.22%
Method: 6, Compressed bytes: 51132.625, Efficiency: 11.23%
Method: 7, Compressed bytes: 56868.5, Efficiency: 1.27%
Method: 8, Compressed bytes: 59625, Efficiency: -3.52%

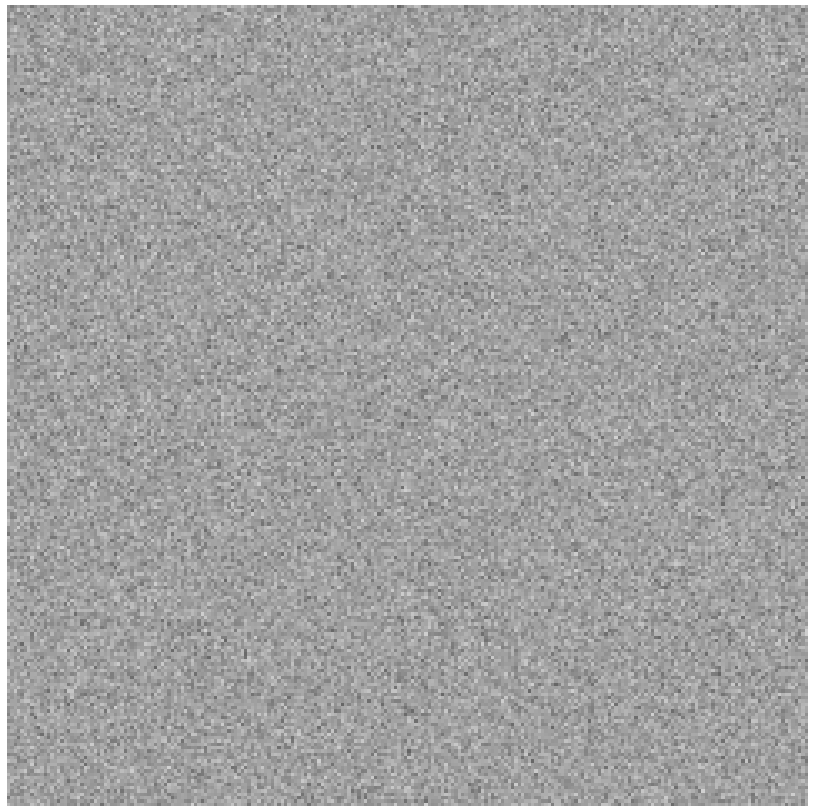
Floating Window:
Method: 1, Compressed bytes: 40598.75, Efficiency: 29.52%
Method: 2, Compressed bytes: 31700.25, Efficiency: 44.96%
Method: 3, Compressed bytes: 35270.875, Efficiency: 38.77%
Method: 4, Compressed bytes: 40422.25, Efficiency: 29.82%
Method: 5, Compressed bytes: 45556.5, Efficiency: 20.91%
Method: 6, Compressed bytes: 50187.125, Efficiency: 12.87%
Method: 7, Compressed bytes: 54837.625, Efficiency: 4.80%

0-Crossing Floating Window:
Method: 1, Compressed bytes: 40598.75, Efficiency: 29.52%
Method: 2, Compressed bytes: 31700.25, Efficiency: 44.96%
Method: 3, Compressed bytes: 35261.25, Efficiency: 38.78%
Method: 4, Compressed bytes: 40307, Efficiency: 30.02%
Method: 5, Compressed bytes: 45543.75, Efficiency: 20.93%
Method: 6, Compressed bytes: 50034.875, Efficiency: 13.13%
Method: 7, Compressed bytes: 54656.875, Efficiency: 5.11%

Adaptive Floating Window:
Method: 1, Compressed bytes: 40598.75, Efficiency: 29.52%
Method: 2, Compressed bytes: 31617.75, Efficiency: 45.11%
Method: 3, Compressed bytes: 35125.625, Efficiency: 39.02%
Method: 4, Compressed bytes: 40217.5, Efficiency: 30.18%
Method: 5, Compressed bytes: 45340.125, Efficiency: 21.28%
Method: 6, Compressed bytes: 49579.375, Efficiency: 13.92%
Method: 7, Compressed bytes: 54517.75, Efficiency: 5.35%

Results

Noise 240x240 Grayscale
Raw bytes: 57600
Max Block Size: 128 bytes
Header bits: 11



Fixed Window:
Method: 0, Compressed bytes: 58218.125, Efficiency: -1.07%
Method: 1, Compressed bytes: 58253.75, Efficiency: -1.13%
Method: 2, Compressed bytes: 58219.25, Efficiency: -1.08%
Method: 3, Compressed bytes: 58392, Efficiency: -1.37%
Method: 4, Compressed bytes: 58567.875, Efficiency: -1.68%
Method: 5, Compressed bytes: 58407.375, Efficiency: -1.40%
Method: 6, Compressed bytes: 53252.75, Efficiency: 7.55%
Method: 7, Compressed bytes: 51468.75, Efficiency: 10.64%
Method: 8, Compressed bytes: 58218.75, Efficiency: -1.07%

Floating Window:
Method: 1, Compressed bytes: 58232.5, Efficiency: -1.10%
Method: 2, Compressed bytes: 58308.5, Efficiency: -1.23%
Method: 3, Compressed bytes: 58213, Efficiency: -1.06%
Method: 4, Compressed bytes: 58573, Efficiency: -1.69%
Method: 5, Compressed bytes: 58958.875, Efficiency: -2.36%
Method: 6, Compressed bytes: 57517.25, Efficiency: 0.14%
Method: 7, Compressed bytes: 52210.625, Efficiency: 9.36%

0-Crossing Floating Window:
Method: 1, Compressed bytes: 58232.5, Efficiency: -1.10%
Method: 2, Compressed bytes: 58308.5, Efficiency: -1.23%
Method: 3, Compressed bytes: 58213, Efficiency: -1.06%
Method: 4, Compressed bytes: 58573, Efficiency: -1.69%
Method: 5, Compressed bytes: 58958.875, Efficiency: -2.36%
Method: 6, Compressed bytes: 57517.25, Efficiency: 0.14%
Method: 7, Compressed bytes: 52210.625, Efficiency: 9.36%

Adaptive Floating Window:
Method: 1, Compressed bytes: 58232.5, Efficiency: -1.10%
Method: 2, Compressed bytes: 58308.5, Efficiency: -1.23%
Method: 3, Compressed bytes: 58213, Efficiency: -1.06%
Method: 4, Compressed bytes: 58573, Efficiency: -1.69%
Method: 5, Compressed bytes: 58958.875, Efficiency: -2.36%
Method: 6, Compressed bytes: 57517.25, Efficiency: 0.14%
Method: 7, Compressed bytes: 52210.625, Efficiency: 9.36%

Results

MIT + Lorem Ipsum
Raw bytes: 32768
Max Block Size: 65536 bytes
Header bits: 20

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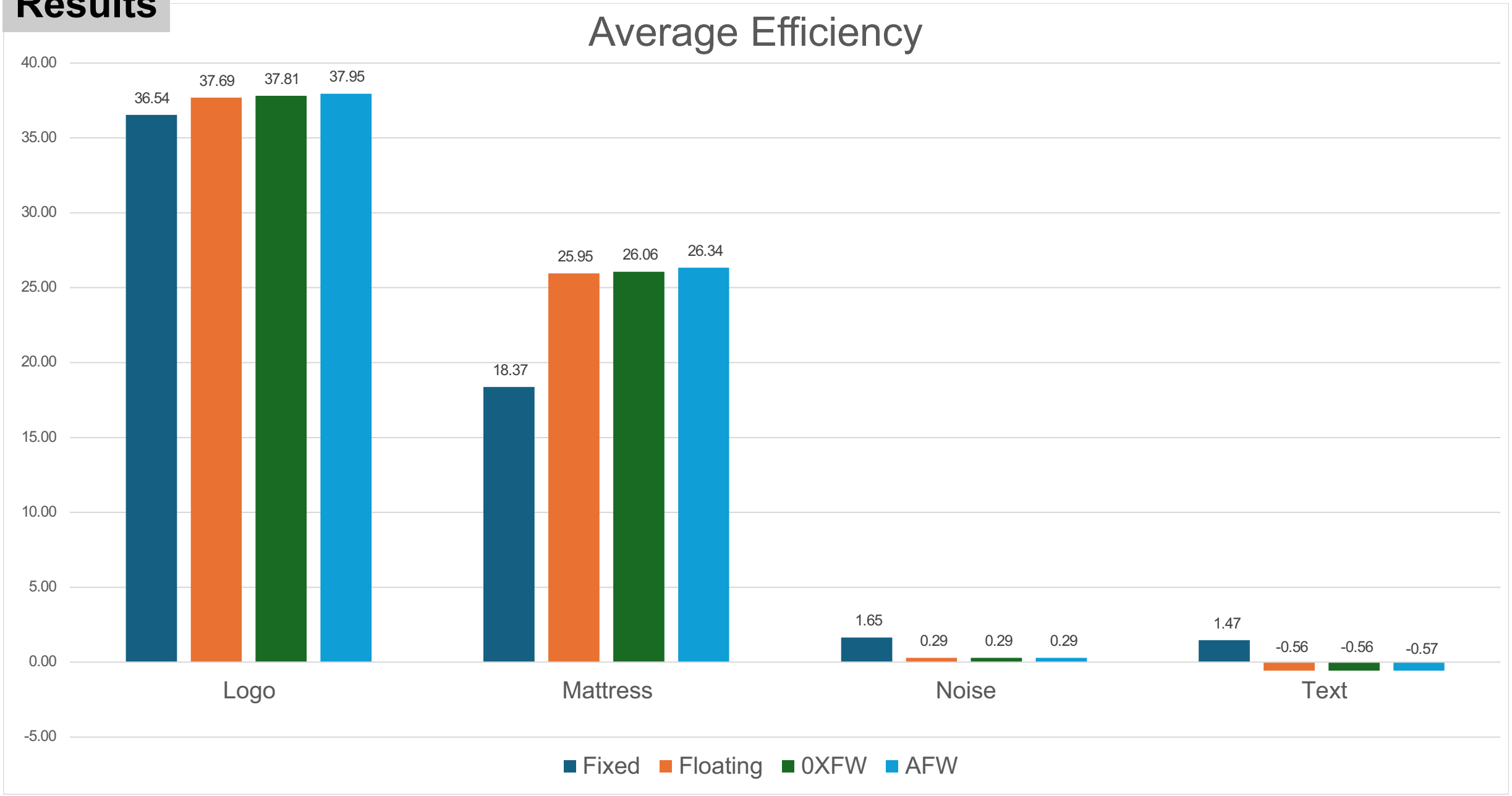
Fixed Window:
Method: 0, Compressed bytes: 32509.5, Efficiency: 0.79%
Method: 1, Compressed bytes: 32544.375, Efficiency: 0.68%
Method: 2, Compressed bytes: 32669.25, Efficiency: 0.30%
Method: 3, Compressed bytes: 32836.875, Efficiency: -0.21%
Method: 4, Compressed bytes: 33440.5, Efficiency: -2.05%
Method: 5, Compressed bytes: 33201, Efficiency: -1.32%
Method: 6, Compressed bytes: 32631.5, Efficiency: 0.42%
Method: 7, Compressed bytes: 28675.5, Efficiency: 12.49%
Method: 8, Compressed bytes: 32770.5, Efficiency: -0.01%

Floating Window:
Method: 1, Compressed bytes: 32544, Efficiency: 0.68%
Method: 2, Compressed bytes: 32578.5, Efficiency: 0.58%
Method: 3, Compressed bytes: 32892.375, Efficiency: -0.38%
Method: 4, Compressed bytes: 33124.5, Efficiency: -1.09%
Method: 5, Compressed bytes: 33701.875, Efficiency: -2.85%
Method: 6, Compressed bytes: 33093.25, Efficiency: -0.99%
Method: 7, Compressed bytes: 32732.5, Efficiency: 0.11%

0-Crossing Floating Window:
Method: 1, Compressed bytes: 32544, Efficiency: 0.68%
Method: 2, Compressed bytes: 32578.5, Efficiency: 0.58%
Method: 3, Compressed bytes: 32892.375, Efficiency: -0.38%
Method: 4, Compressed bytes: 33124.5, Efficiency: -1.09%
Method: 5, Compressed bytes: 33701.875, Efficiency: -2.85%
Method: 6, Compressed bytes: 33093.25, Efficiency: -0.99%
Method: 7, Compressed bytes: 32732.5, Efficiency: 0.11%

Adaptive Floating Window:
Method: 1, Compressed bytes: 32544, Efficiency: 0.68%
Method: 2, Compressed bytes: 32578.5, Efficiency: 0.58%
Method: 3, Compressed bytes: 32892.375, Efficiency: -0.38%
Method: 4, Compressed bytes: 33124.5, Efficiency: -1.09%
Method: 5, Compressed bytes: 33701.875, Efficiency: -2.85%
Method: 6, Compressed bytes: 33087.25, Efficiency: -0.97%
Method: 7, Compressed bytes: 32752.875, Efficiency: 0.05%

Results


Average Efficiency
40.00
35.00
30.00
25.00
20.00
15.00
10.00
5.00
0.00
-5.00
36.54
37.69
37.81
37.95
18.37
25.95
26.06
26.34
1.65
0.29
0.29
0.29
1.47
-0.56
-0.56
-0.57
Logo
Mattress
Noise
Text
Fixed
Floating
0XFW
AFW

Next sprint

- Finding a way to build an effective sequence of blocks.